

2013
**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I Pages 2–6

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 7–15

90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 What are the solutions of $2x^2 - 5x - 1 = 0$?

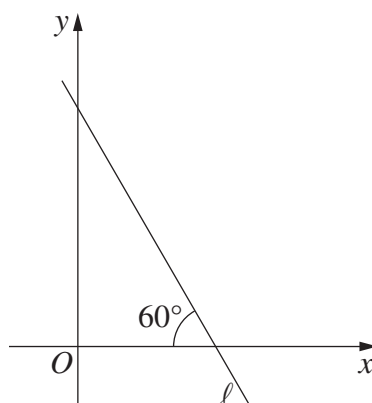
(A) $x = \frac{-5 \pm \sqrt{17}}{4}$

(B) $x = \frac{5 \pm \sqrt{17}}{4}$

(C) $x = \frac{-5 \pm \sqrt{33}}{4}$

(D) $x = \frac{5 \pm \sqrt{33}}{4}$

2 The diagram shows the line ℓ .



What is the slope of the line ℓ ?

(A) $\sqrt{3}$

(B) $-\sqrt{3}$

(C) $\frac{1}{\sqrt{3}}$

(D) $-\frac{1}{\sqrt{3}}$

3 Which inequality defines the domain of the function $f(x) = \frac{1}{\sqrt{x+3}}$?

(A) $x > -3$

(B) $x \geq -3$

(C) $x < -3$

(D) $x \leq -3$

4 What is the derivative of $\frac{x}{\cos x}$?

(A) $\frac{\cos x + x \sin x}{\cos^2 x}$

(B) $\frac{\cos x - x \sin x}{\cos^2 x}$

(C) $\frac{x \sin x - \cos x}{\cos^2 x}$

(D) $\frac{-x \sin x - \cos x}{\cos^2 x}$

5 A bag contains 4 red marbles and 6 blue marbles. Three marbles are selected at random without replacement.

What is the probability that at least one of the marbles selected is red?

(A) $\frac{1}{6}$

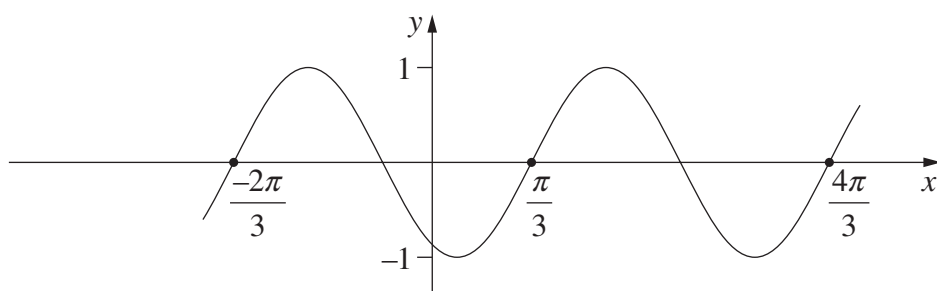
(B) $\frac{1}{2}$

(C) $\frac{5}{6}$

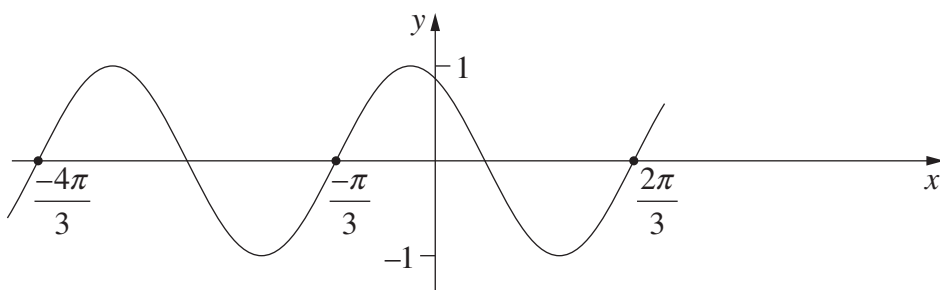
(D) $\frac{29}{30}$

6 Which diagram shows the graph $y = \sin\left(2x + \frac{\pi}{3}\right)$?

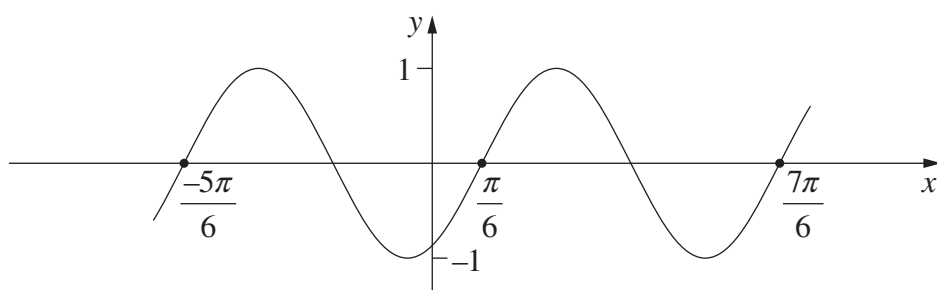
(A)



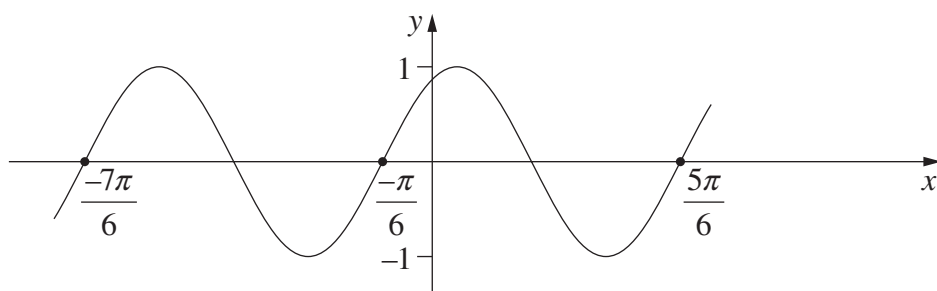
(B)



(C)



(D)

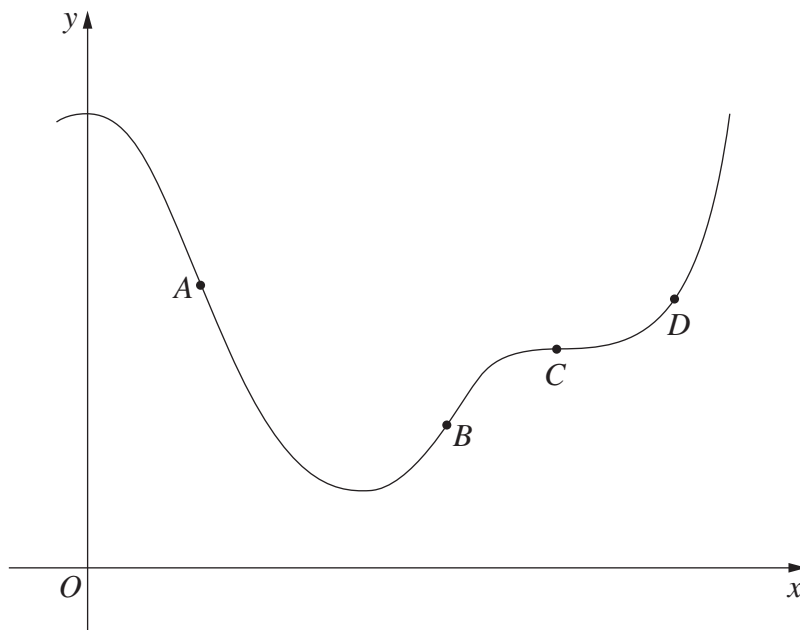


- 7 A parabola has focus $(5, 0)$ and directrix $x = 1$.

What is the equation of the parabola?

- (A) $y^2 = 16(x - 5)$
- (B) $y^2 = 8(x - 3)$
- (C) $y^2 = -16(x - 5)$
- (D) $y^2 = -8(x - 3)$

- 8 The diagram shows points A , B , C and D on the graph $y = f(x)$.



At which point is $f'(x) > 0$ and $f''(x) = 0$?

- (A) A
- (B) B
- (C) C
- (D) D

9 What is the solution of $5^x = 4$?

(A) $x = \frac{\log_e 4}{5}$

(B) $x = \frac{4}{\log_e 5}$

(C) $x = \frac{\log_e 4}{\log_e 5}$

(D) $x = \log_e \left(\frac{4}{5} \right)$

10 A particle is moving along the x -axis. The displacement of the particle at time t seconds is x metres.

At a certain time, $\dot{x} = -3 \text{ m s}^{-1}$ and $\ddot{x} = 2 \text{ m s}^{-2}$.

Which statement describes the motion of the particle at that time?

- (A) The particle is moving to the right with increasing speed.
- (B) The particle is moving to the left with increasing speed.
- (C) The particle is moving to the right with decreasing speed.
- (D) The particle is moving to the left with decreasing speed.

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet.

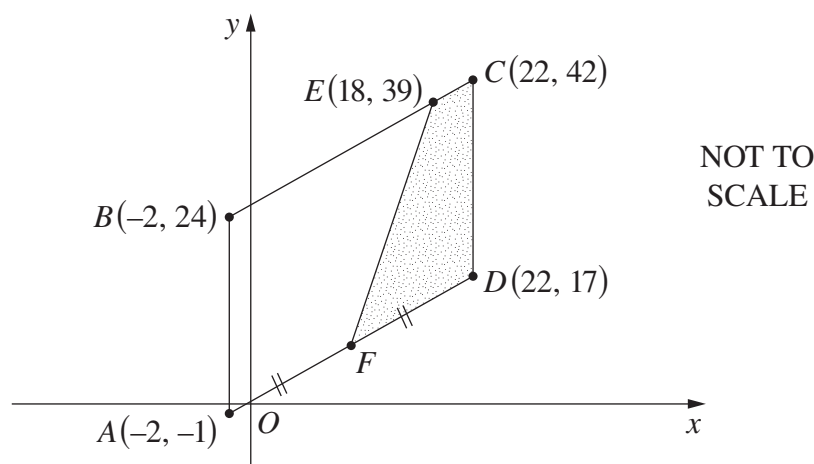
- (a) Evaluate $\ln 3$ correct to three significant figures. **1**
- (b) Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4}$. **2**
- (c) Differentiate $(\sin x - 1)^8$. **2**
- (d) Differentiate $x^2 e^x$. **2**
- (e) Find $\int e^{4x+1} dx$. **2**
- (f) Evaluate $\int_0^1 \frac{x^2}{x^3 + 1} dx$. **3**
- (g) Sketch the region defined by $(x - 2)^2 + (y - 3)^2 \geq 4$. **3**

Question 12 (15 marks) Use the Question 12 Writing Booklet.

- (a) The cubic $y = ax^3 + bx^2 + cx + d$ has a point of inflexion at $x = p$. 2

Show that $p = -\frac{b}{3a}$.

- (b) The points $A(-2, -1)$, $B(-2, 24)$, $C(22, 42)$ and $D(22, 17)$ form a parallelogram as shown. The point $E(18, 39)$ lies on BC . The point F is the midpoint of AD .



- (i) Show that the equation of the line through A and D is $3x - 4y + 2 = 0$. 2
- (ii) Show that the perpendicular distance from B to the line through A and D is 20 units. 1
- (iii) Find the length of EC . 1
- (iv) Find the area of the trapezium $EFDC$. 2
- (c) Kim and Alex start jobs at the beginning of the same year. Kim's annual salary in the first year is \$30 000, and increases by 5% at the beginning of each subsequent year. Alex's annual salary in the first year is \$33 000, and increases by \$1500 at the beginning of each subsequent year.
- (i) Show that in the 10th year Kim's annual salary is higher than Alex's annual salary. 2
- (ii) In the first 10 years how much, in total, does Kim earn? 2
- (iii) Every year, Alex saves $\frac{1}{3}$ of her annual salary. How many years does it take her to save \$87 500? 3

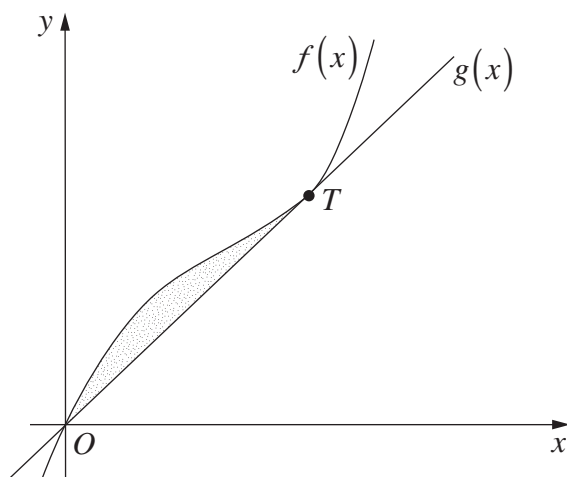
Question 13 (15 marks) Use the Question 13 Writing Booklet.

- (a) The population of a herd of wild horses is given by

$$P(t) = 400 + 50 \cos\left(\frac{\pi}{6}t\right),$$

where t is time in months.

- (i) Find all times during the first 12 months when the population equals 375 horses. **2**
- (ii) Sketch the graph of $P(t)$ for $0 \leq t \leq 12$. **2**
- (b) The diagram shows the graphs of the functions $f(x) = 4x^3 - 4x^2 + 3x$ and $g(x) = 2x$. The graphs meet at O and at T .



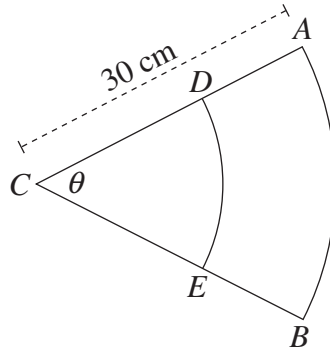
- (i) Find the x -coordinate of T . **1**
- (ii) Find the area of the shaded region between the graphs of the functions $f(x)$ and $g(x)$. **3**

Question 13 continues on page 10

Question 13 (continued)

- (c) The region ABC is a sector of a circle with radius 30 cm, centred at C . The angle of the sector is θ . The arc DE lies on a circle also centred at C , as shown in the diagram.

2



The arc DE divides the sector ABC into two regions of equal area.

Find the exact length of the interval CD .

- (d) A family borrows \$500 000 to buy a house. The loan is to be repaid in equal monthly instalments. The interest, which is charged at 6% per annum, is reducible and calculated monthly. The amount owing after n months, A_n , is given by

$$A_n = Pr^n - M(1 + r + r^2 + \dots + r^{n-1}), \quad (\text{Do NOT prove this})$$

where P is the amount borrowed, $r = 1.005$ and M is the monthly repayment.

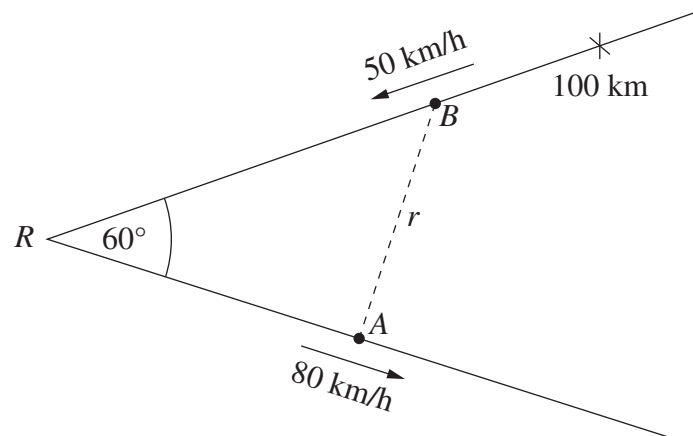
- (i) The loan is to be repaid over 30 years. Show that the monthly repayment is \$2998 to the nearest dollar.
- (ii) Show that the balance owing after 20 years is \$270 000 to the nearest thousand dollars.
- (iii) After 20 years the family borrows an extra amount, so that the family then owes a total of \$370 000. The monthly repayment remains \$2998, and the interest rate remains the same.

How long will it take to repay the \$370 000?

End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet.

- (a) The velocity of a particle moving along the x -axis is given by $\dot{x} = 10 - 2t$, where x is the displacement from the origin in metres and t is the time in seconds. Initially the particle is 5 metres to the right of the origin.
- (i) Show that the acceleration of the particle is constant. 1
 - (ii) Find the time when the particle is at rest. 1
 - (iii) Show that the position of the particle after 7 seconds is 26 metres to the right of the origin. 2
 - (iv) Find the distance travelled by the particle during the first 7 seconds. 2
- (b) Two straight roads meet at R at an angle of 60° . At time $t = 0$ car A leaves R on one road, and car B is 100 km from R on the other road. Car A travels away from R at a speed of 80 km/h, and car B travels towards R at a speed of 50 km/h.



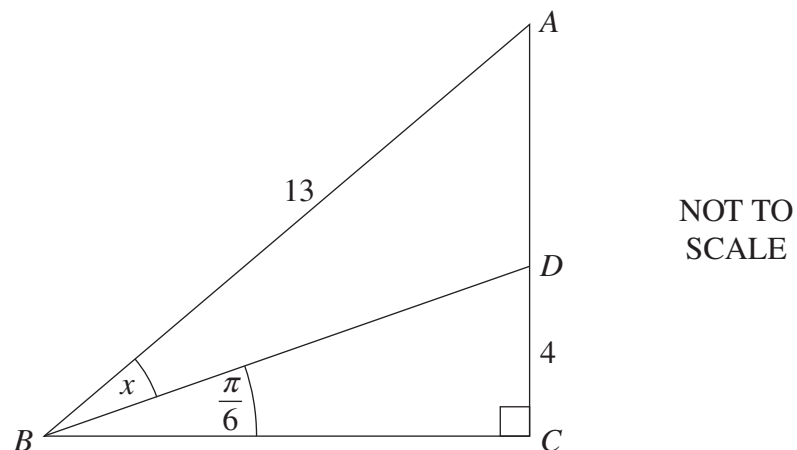
The distance between the cars at time t hours is r km.

- (i) Show that $r^2 = 12\,900t^2 - 18\,000t + 10\,000$. 2
- (ii) Find the minimum distance between the cars. 3

Question 14 continues on page 12

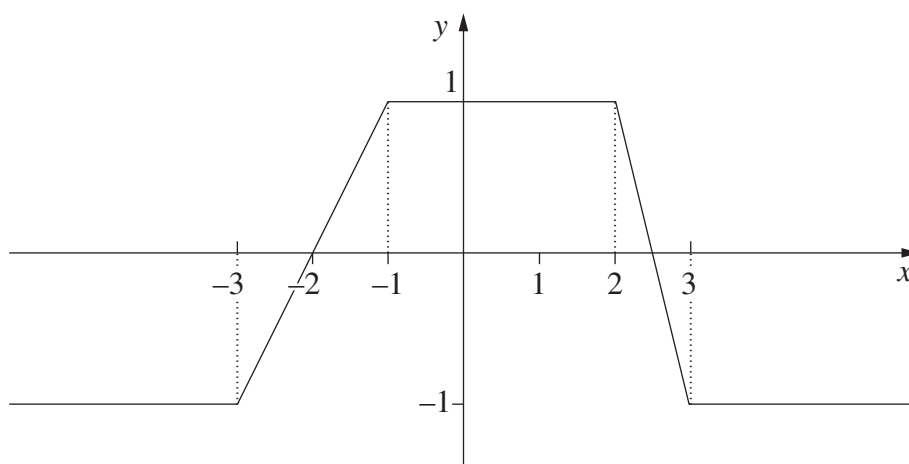
Question 14 (continued)

- (c) The right-angled triangle ABC has hypotenuse $AB = 13$. The point D is on AC such that $DC = 4$, $\angle DBC = \frac{\pi}{6}$ and $\angle ABD = x$. **3**



Using the sine rule, or otherwise, find the exact value of $\sin x$.

- (d) The diagram shows the graph $y = f(x)$. **1**

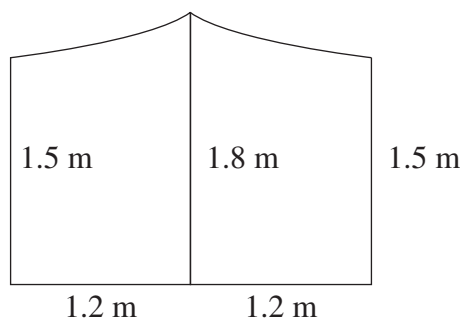


What is the value of a , where $a > 0$, so that $\int_{-a}^a f(x) dx = 0$?

End of Question 14

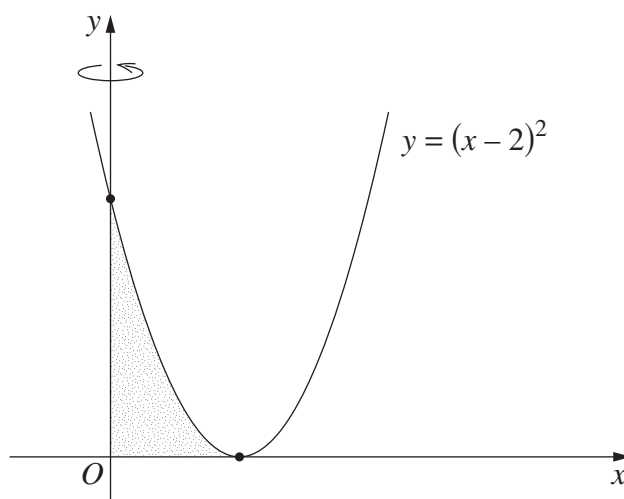
Question 15 (15 marks) Use the Question 15 Writing Booklet.

- (a) The diagram shows the front of a tent supported by three vertical poles. The poles are 1.2 m apart. The height of each outer pole is 1.5 m, and the height of the middle pole is 1.8 m. The roof hangs between the poles.



The front of the tent has area $A \text{ m}^2$.

- (i) Use the trapezoidal rule to estimate A . **1**
- (ii) Use Simpson's rule to estimate A . **1**
- (iii) Explain why the trapezoidal rule gives the better estimate of A . **1**
- (b) The region bounded by the x -axis, the y -axis and the parabola $y = (x - 2)^2$ is rotated about the y -axis to form a solid. **4**



Find the volume of the solid.

Question 15 continues on page 14

Question 15 (continued)

- (c) (i) Sketch the graph $y = |2x - 3|$. **1**
- (ii) Using the graph from part (i), or otherwise, find all values of m for which the equation $|2x - 3| = mx + 1$ has exactly one solution. **2**
- (d) Pat and Chandra are playing a game. They take turns throwing two dice. The game is won by the first player to throw a double six. Pat starts the game.
- (i) Find the probability that Pat wins the game on the first throw. **1**
- (ii) What is the probability that Pat wins the game on the first or on the second throw? **2**
- (iii) Find the probability that Pat eventually wins the game. **2**

End of Question 15

Question 16 (15 marks) Use the Question 16 Writing Booklet.

- (a) The derivative of a function $f(x)$ is $f'(x) = 4x - 3$. The line $y = 5x - 7$ is tangent to the graph of $f(x)$. **3**

Find the function $f(x)$.

- (b) Trout and carp are types of fish. A lake contains a number of trout. At a certain time 10 carp are introduced into the lake and start eating the trout. As a consequence, the number of trout, N , decreases according to

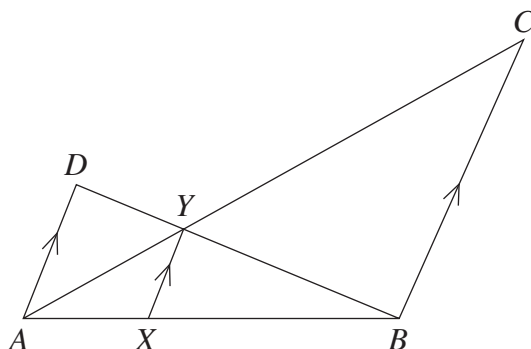
$$N = 375 - e^{0.04t},$$

where t is the time in months after the carp are introduced.

The population of carp, P , increases according to

$$\frac{dP}{dt} = 0.02P.$$

- (i) How many trout were in the lake when the carp were introduced? **1**
- (ii) When will the population of trout be zero? **1**
- (iii) Sketch the number of trout as a function of time. **1**
- (iv) When is the rate of increase of carp equal to the rate of decrease of trout? **3**
- (v) When is the number of carp equal to the number of trout? **2**
- (c) The diagram shows triangles ABC and ABD with AD parallel to BC . The sides AC and BD intersect at Y . The point X lies on AB such that XY is parallel to AD and BC .



- (i) Prove that $\triangle ABC$ is similar to $\triangle AXY$. **2**
- (ii) Hence, or otherwise, prove that $\frac{1}{XY} = \frac{1}{AD} + \frac{1}{BC}$. **2**

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



2013 HSC Mathematics Marking Guidelines

Section I

Multiple-choice Answer Key

Question	Answer
1	D
2	B
3	A
4	A
5	C
6	D
7	B
8	B
9	C
10	D

Section II

Question 11 (a)

Criteria	Marks
• Correct answer	1

Sample answer:

$$\ln 3 = 1.09861\dots$$

$$= 1.10 \text{ (to 3 significant figures)}$$

Question 11 (b)

Criteria	Marks
• Correct solution	2
• Attempts to factorise numerator and denominator, or equivalent merit	1

Sample answer:

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4} &= \lim_{x \rightarrow 2} \frac{(x - 2)(x^2 + 2x + 4)}{(x - 2)(x + 2)} \\ &= \lim_{x \rightarrow 2} \frac{x^2 + 2x + 4}{x + 2} \\ &= \frac{2^2 + 2 \times 2 + 4}{2 + 2} = 3 \end{aligned}$$

Question 11 (c)

Criteria	Marks
• Correct derivative	2
• Attempts to use chain rule, or equivalent merit	1

Sample answer:

$$\frac{d}{dx}(\sin x - 1)^8 = 8(\sin x - 1)^7 \cos x$$

Question 11 (d)

Criteria	Marks
• Correct derivative	2
• Attempts to use the product rule	1

Sample answer:

$$\frac{d}{dx}(x^2 e^x) = 2xe^x + x^2 e^x$$

Question 11 (e)

Criteria	Marks
• Correct primitive	2
• Obtains primitive of the form ae^{4x+1}	1

Sample answer:

$$\int e^{4x+1} dx = \frac{1}{4}e^{4x+1} + C$$

Question 11 (f)

Criteria	Marks
• Correct solution	3
• Obtains correct primitive OR • Substitutes limits into primitive of the form $a \ln(x^3 + 1)$, or equivalent merit	2
• Obtains primitive of the form $a \ln(x^3 + 1)$, or equivalent merit	1

Sample answer:

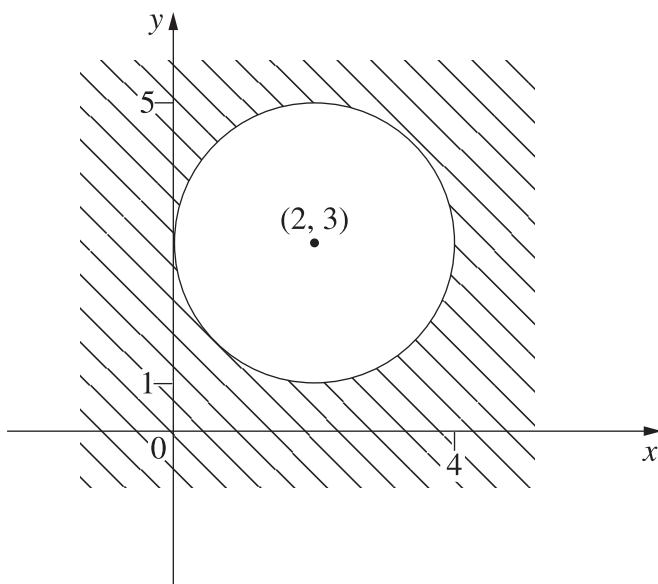
$$\begin{aligned}
 \int_0^1 \frac{x^2}{x^3 + 1} dx &= \frac{1}{3} \int_0^1 \frac{3x^2}{x^3 + 1} dx \\
 &= \frac{1}{3} \left[\ln(x^3 + 1) \right]_0^1 \\
 &= \frac{1}{3} \left[\ln(1^3 + 1) - \ln(0^3 + 1) \right] \\
 &= \frac{1}{3} \ln 2
 \end{aligned}$$

Question 11 (g)

Criteria	Marks
• Correct sketch	3
• Shows circle with TWO of the following correct: radius, centre, shading	2
• Shows circle with correct radius OR correct centre OR correct shading	1

Sample answer:

Exterior of a circle centre at (2, 3) and a radius 2.



Question 12 (a)

Criteria	Marks
• Correct solution	2
• Finds correct second derivative, or equivalent merit	1

Sample answer:

For a point of inflexion

$$y'' = 0 \text{ at } x = p$$

$$y' = 3ax^2 + 2bx + c$$

$$y'' = 6ax + 2b$$

$$\text{Hence, } 6ax + 2b = 0$$

$$\therefore x = \frac{-2b}{6a}$$

$$\text{at } x = p, \quad p = \frac{-b}{3a}$$

Question 12 (b) (i)

Criteria	Marks
• Correct solution	2
• Finds correct slope, or equivalent merit	1

Sample answer:

$$\begin{aligned}\text{Slope of the line is } \frac{17 - (-1)}{22 - (-2)} &= \frac{18}{24} \\ &= \frac{3}{4}\end{aligned}$$

Since $(-2, -1)$ is on the line, the equation of the line is

$$\begin{aligned}y - (-1) &= \frac{3}{4}(x - (-2)) \\ 4y + 4 &= 3x + 6 \\ 0 &= 3x - 4y + 2\end{aligned}$$

Question 12 (b) (ii)

Criteria	Marks
• Correct solution	1

Sample answer:

Using the perpendicular distance formula,

$$\begin{aligned}d &= \frac{|3(-2) - 4 \times 24 + 2|}{\sqrt{3^2 + 4^2}} \\ &= \frac{100}{5} \\ &= 20\end{aligned}$$

$\therefore$ the distance is 20 units

Question 12 (b) (iii)

Criteria	Marks
• Correct solution	1

Sample answer:

$$\begin{aligned}
 EC &= \sqrt{(42 - 39)^2 + (22 - 18)^2} \\
 &= \sqrt{3^2 + 4^2} \\
 &= 5
 \end{aligned}$$

∴ the length of EC is 5 units

Question 12 (b) (iv)

Criteria	Marks
• Correct solution	2
• Finds the length of FD , or equivalent merit	1

Sample answer:

$$\begin{aligned}
 AD &= \sqrt{(22 - (-21))^2 + (17 - (-11))^2} \\
 &= \sqrt{24^2 + 18^2} \\
 &= 30
 \end{aligned}$$

Since F is the midpoint of AD , $FD = 15$

The height of the trapezium is the perpendicular distance from part (ii).

$$\begin{aligned}
 \therefore \text{Area of trapezium} &= \frac{1}{2} \times 20 \times (5 + 15) \text{ units}^2 \\
 &= 200 \text{ units}^2
 \end{aligned}$$

Question 12 (c) (i)

Criteria	Marks
• Correct answer	2
• Obtains Kim's or Alex's salary after 10 years, or equivalent merit	1

Sample answer:

Kim's salary is a geometric progression where $a = 30\,000$ and $r = 1.05$

$$\begin{aligned} T_{10} &= 30\,000 + (1.05)^{10-1} \\ &= 46\,539.85 \end{aligned}$$

Alex's salary is an arithmetic progression where $a = 33\,000$ and $d = 1500$

$$\begin{aligned} T_{10} &= 33\,000 + (10 - 1) \times 1\,500 \\ &= 46\,500 \end{aligned}$$

Hence Kim's salary is higher than Alex's salary in the 10th year.

Question 12 (c) (ii)

Criteria	Marks
• Correct solution	2
• Attempts to find the sum of a relevant geometric series, or equivalent merit	1

Sample answer:

Sum of a geometric progression

$$\begin{aligned} S_{10} &= \frac{30\,000(1.05^{10} - 1)}{1.05 - 1} \\ &= 377\,336.78 \end{aligned}$$

Kim earned \$377 336.78 in the first 10 years.

Question 12 (c) (iii)

Criteria	Marks
• Correct solution	3
• Attempts to solve resulting quadratic equation	2
• Attempts to use the sum of an arithmetic series to get a quadratic equation or equivalent merit	1

Sample answer:

Alex saves

11 000 in the first year
 11 000 + 500 in the second year
 11 000 + 500 + 500 in the third year
 ⋮
 $11\,000 + \underbrace{500 + \dots + 500}_{n-1}$ in the n^{th} year
 (ie = $11\,000 + (n-1)500$ in the n^{th} year)

Total savings in n years:

$$\begin{aligned}
 S_n &= \frac{n}{2}(2a + (n-1)d) \\
 S_n &= \frac{n}{2}(2(11\,000) + (n-1)500) \\
 &= \frac{n}{2}(22\,000 + 500n - 500) \\
 &= \frac{n}{2}(21\,500 + 500n)
 \end{aligned}$$

Now $S_n = 87\,500$

$$21\,500n + 500n^2 = 2 \times (87\,500) = 175\,000$$

$$500n^2 + 21\,500n - 175\,000 = 0$$

$$n^2 + 43n - 350 = 0$$

$$(n+50)(n-7) = 0$$

so $n = 7$ or $n = -50$

But years must be positive, so Alex will save \$87 500 after seven years.

Question 13 (a) (i)

Criteria	Marks
• Correct solution	2
• Obtains one of the times, or equivalent merit	1

Sample answer:

$$P(t) = 400 + 50 \cos \frac{\pi}{6} t = 375$$

$$50 \cos \frac{\pi}{6} t = -25$$

$$\cos \frac{\pi}{6} t = -\frac{1}{2}$$

$$\text{Hence } \frac{\pi}{6} t = \frac{2\pi}{3}, \frac{4\pi}{3}, \dots$$

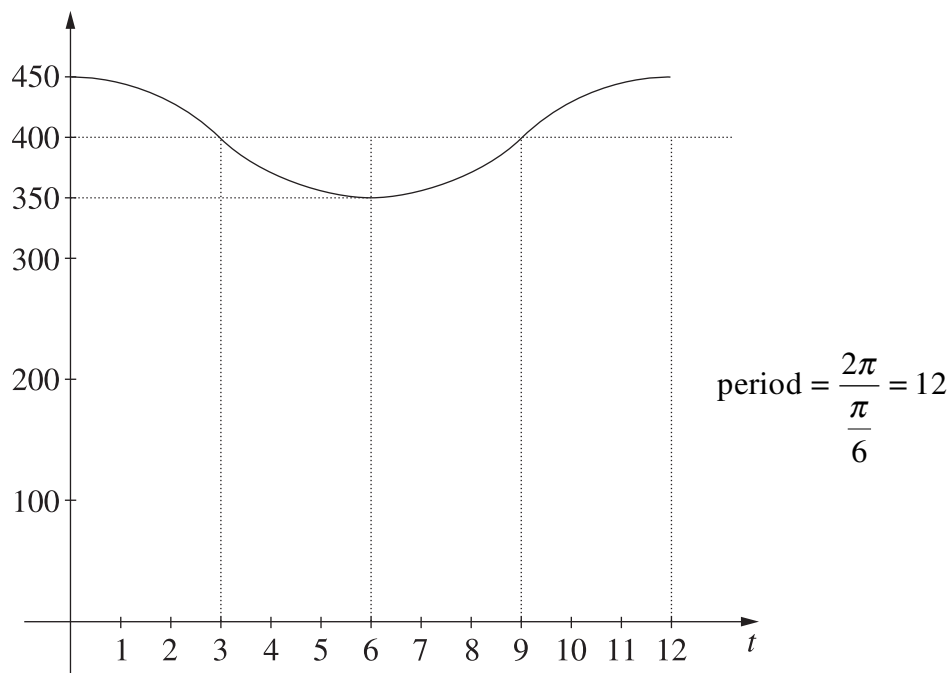
$$t = 4, 8$$

The population is 375 at 4 and at 8 months.

Question 13 (a) (ii)

Criteria	Marks
• Correct graph	2
• Shows a periodic function with two of the following features correct: period, amplitude, phase, centre of motion	1

Sample answer:



Question 13 (b) (i)

Criteria	Marks
• Correct solution	1

Sample answer:

x - coordinate of T satisfies $f(x) = g(x)$

$$4x^3 - 4x^2 + 3x = 2x$$

$$4x^3 - 4x^2 + x = 0$$

$$x(4x^2 - 4x + 1) = 0$$

$$x(2x - 1)^2 = 0$$

$$\text{Hence } x = 0 \text{ or } x = \frac{1}{2}$$

As T is not the origin, $x = \frac{1}{2}$.

Question 13 (b) (ii)

Criteria	Marks
• Correct solution	3
• Obtains correct primitive, or equivalent merit	2
• Obtains a definite integral for the area, or equivalent merit	1

Sample answer:

The area is given by

$$\int_0^{\frac{1}{2}} (4x^3 - 4x^2 + 3x - 2x) dx$$

$$= \int_0^{\frac{1}{2}} (4x^3 - 4x^2 + x) dx$$

$$= \left[\frac{4x^4}{4} - \frac{4x^3}{3} + \frac{x^2}{2} \right]_0^{\frac{1}{2}}$$

$$= \frac{1}{48}$$

$$\therefore \text{Area} = \frac{1}{48} \text{ units}^2$$

Question 13 (c)

Criteria	Marks
• Correct solution	2
• Finds the area of the sector ABC in terms of θ , or equivalent merit	1

Sample answer:

Area of sector ABC is $\frac{1}{2} \times 30^2 \theta$

Area of sector DEC is half of that:

$$\frac{1}{4} \times 30^2 \theta = 225\theta$$

$$\text{Hence } \frac{1}{2}(CD)^2 \theta = 225\theta$$

$$\begin{aligned}(CD)^2 &= 450 \\ CD &= \sqrt{450} \\ &= 15\sqrt{2}\end{aligned}$$

Length of CD is $15\sqrt{2}$ cm.

Question 13 (d) (i)

Criteria	Marks
• Correct solution	2
• Substitutes given information into $A_{360} = 0$, or equivalent merit	1

Sample answer:

M needs to be such that

$$\begin{aligned}0 &= 500\,000(1.005)^{360} - M(1 + 1.005 + \dots + (1.005)^{359}) \\ &= 500\,000(1.005)^{360} - M \frac{(1.005)^{360} - 1}{1.005 - 1} \\ &= 500\,000(1.005)^{360} - 1004.515 M\end{aligned}$$

Hence

$$\begin{aligned}M &= \frac{500\,000(1.005)^{360}}{1004.515} \\ &= 2997.75\end{aligned}$$

$\therefore$ monthly repayment is = \$2998 to the nearest dollar.

Question 13 (d) (ii)

Criteria	Marks
• Correct solution	1

Sample answer:

$$\begin{aligned}
 A_{240} &= 500\,000(1.005)^{240} - 2998(1 + 1.005 + \dots + (1.005)^{239}) \\
 &= 500\,000(1.005)^{240} - 2998 \frac{(1.005)^{240} - 1}{1.005 - 1} \\
 &= 269\,903.63
 \end{aligned}$$

∴ Balance owing is \$270 000 to the nearest thousand.

Question 13 (d) (iii)

Criteria	Marks
• Correct solution	2
• Finds equation in terms of $(1.005)^n$, or equivalent merit	1

Sample answer:

Need to find n so that

$$370\,000(1.005)^n - 2998(1 + 1.005 + \dots + (1.005)^{n-1}) = 0$$

$$370\,000(1.005)^n = 2998 \frac{(1.005)^n - 1}{1.005 - 1}$$

$$370\,000(1.005)^n = 599\,600((1.005)^n - 1)$$

$$599\,600 = (599\,600 - 370\,000)(1.005)^n$$

$$599\,600 = 229\,600(1.005)^n$$

$$2.6115 = (1.005)^n$$

$$\ln(2.6115) = n \ln(1.005)$$

$$n = \frac{\ln(2.6115)}{\ln(1.005)}$$

$$= 192.46$$

Hence 193 months (that is, 16 years and 1 month) are required to repay the \$370 000.

Question 14 (a) (i)

Criteria	Marks
• Correct solution	1

Sample answer:

Acceleration is $\ddot{x}$

$$\dot{x} = 10 - 2t$$

$$\ddot{x} = -2$$

$\therefore$ acceleration is constant.

Question 14 (a) (ii)

Criteria	Marks
• Correct solution	1

Sample answer:

At rest when $\dot{x} = 0$

$$\dot{x} = 10 - 2t = 0$$

$$t = 5$$

The particle is at rest after 5 seconds.

Question 14 (a) (iii)

Criteria	Marks
• Correct solution	2
• Finds $x = 10t - t^2 + C$, or equivalent merit	1

Sample answer:

Displacement of particle is

$$\begin{aligned}
 x &= \int (10 - 2t) dt \\
 &= 10t - t^2 + C
 \end{aligned}$$

When $t = 0$, $x = 5$

$$x = C = 5$$

$$\text{ie } x = 10t - t^2 + 5.$$

After 7 seconds,

$$\begin{aligned}
 x &= 10 \times 7 - 7^2 + 5 \\
 &= 26
 \end{aligned}$$

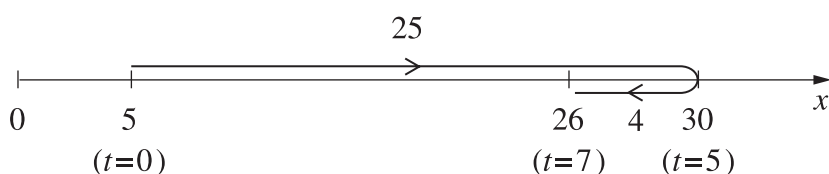
$\therefore$ particle is 26 m to the right of the origin.

Question 14 (a) (iv)

Criteria	Marks
• Correct solution	2
• Finds displacement when particle is at rest, or equivalent merit	1

Sample answer:

The particle is at rest when $t = 5$, that is, $x = 30$. Hence particle travels from $x = 5$ to $x = 30$, then to $x = 26$. Total distance : 25 m + 4 m = 29 m



Question 14 (b) (i)

Criteria	Marks
• Correct solution	2
• Finds BR in terms of t , or equivalent merit	1

Sample answer:

After time t , distance of A to R is $80t$

After time t , distance of B to R is $100 - 50t$

Using the cosine rule

$$\begin{aligned}
 D^2 &= (100 - 50t)^2 + (80t)^2 - 2(100 - 50t)80t \cos 60^\circ \\
 &= 10\,000 - 10\,000t + 2\,500t^2 + 6\,400t^2 - 160t(100 - 50t)\left(\frac{1}{2}\right) \\
 &= 10\,000 - 10\,000t + 8\,900t^2 - 8\,000t + 4\,000t^2 \\
 &= 12\,900t^2 - 18\,000t + 10\,000
 \end{aligned}$$

Question 14 (b) (ii)

Criteria	Marks
• Correct solution	3
• Justifies that the value of t gives a minimum distance	2
• Finds the time when the distance is minimal, or equivalent merit	1

Sample answer:

Minimum of D occurs at the same time as the minimum of D^2 . For maximum/minimum

$$\frac{d(D^2)}{dt} = 25\,800t - 18\,000$$

$$= 0$$

$$t = 0.69767\dots$$

$$= 0.7 \text{ (to 1 dp)}$$

$$D^2 = 12\,900(0.7)^2 - 18\,000(0.7) + 10\,000$$

$$= 3721$$

$$\therefore D = 61$$

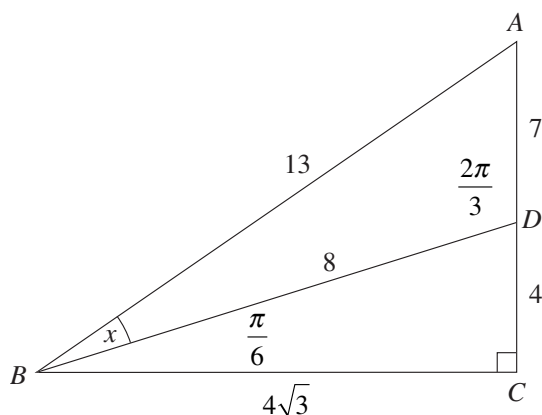
Test: $\frac{d^2(D^2)}{dt^2} = 25\,800 > 0$, so minimum

$\therefore$ Minimum distance is 61km.

Question 14 (c)

Criteria	Marks
• Correct solution	3
• States $\frac{\sin x}{AD} = \frac{\sqrt{3}}{26}$ and attempts to find AD , or equivalent merit	2
• States $\frac{\sin x}{AD} = \frac{\sin \frac{2\pi}{3}}{13}$, or equivalent merit	1

Sample answer:



$$\sin \frac{\pi}{6} = \frac{CD}{BD}$$

$$\frac{1}{2} = \frac{4}{BD}$$

$$\therefore BD = 8$$

$$\tan \frac{\pi}{6} = \frac{CD}{BC}$$

$$\frac{1}{\sqrt{3}} = \frac{4}{BC}$$

$$\therefore BC = 4\sqrt{3}$$

$$\begin{aligned} \text{Now } AC &= \sqrt{AB^2 - BC^2} \\ &= \sqrt{13^2 - (4\sqrt{3})^2} \\ &= \sqrt{121} = 11 \end{aligned}$$

$$\begin{aligned} \text{Hence } AD &= AC - DC \\ &= 11 - 4 = 7 \end{aligned}$$

$$\text{Also } \angle ADB = \frac{\pi}{2} + \frac{\pi}{6} = \frac{2\pi}{3} \text{ (ext. angle of } \triangle BCD)$$

Using the sine rule in $\triangle ABD$

$$\frac{\sin x}{AD} = \frac{\sin \frac{2\pi}{3}}{AB}$$

$$\frac{\sin x}{7} = \frac{\sin \frac{2\pi}{3}}{13}$$

$$\begin{aligned} &= \frac{\sqrt{3}}{13} \\ &= \frac{2}{13} \\ &= \frac{\sqrt{3}}{26} \end{aligned}$$

$$\text{Hence } \sin x = \frac{7\sqrt{3}}{26}$$

Question 14 (d)

Criteria	Marks
• Correct solution	1

Sample answer:

$$a = 4.5$$

Reason:

$$\int_{-3}^{-1} f(x) dx = 0 \text{ and } \int_2^3 f(x) dx = 0$$

$$\begin{aligned} \text{and so } \int_{-3}^3 f(x) dx &= \int_{-1}^2 f(x) dx \\ &= 3 \times 1 \\ &= 3 \end{aligned}$$

So for

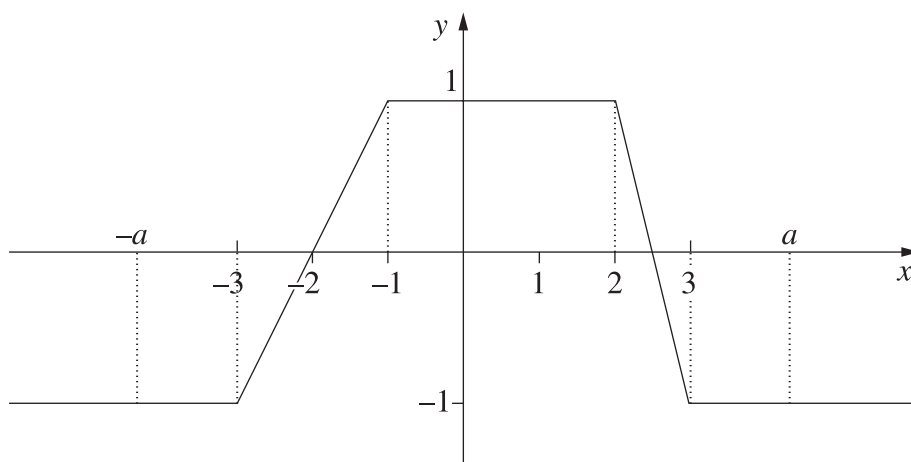
$$\int_{-a}^a f(x) dx = 0$$

$$\int_{-a}^{-3} f(x) dx + \int_3^a f(x) dx = 3$$

$$\text{ie } 1 \times 1.5 + 1 \times 1.5 = 3$$

$$\therefore a = 3 + 1.5$$

$$= 4.5$$



Question 15 (a) (i)

Criteria	Marks
• Correct solution	1

Sample answer:

Estimated area from trapezoidal rule

$$\frac{1.2}{2}(1.5 + 1.5 + 2(1.8)) = 3.96 \text{ m}^2$$

Question 15 (a) (ii)

Criteria	Marks
• Correct solution	1

Sample answer:

Estimated area from Simpson's rule

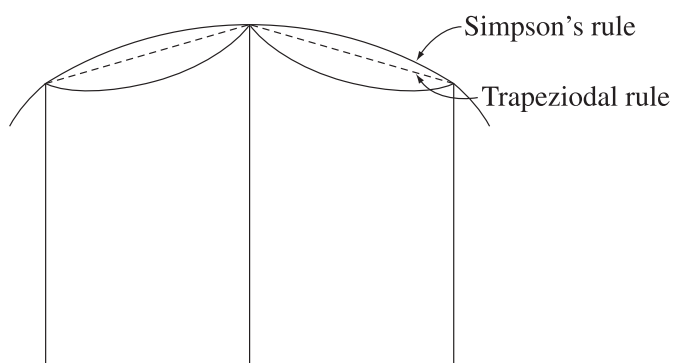
$$\frac{1.2}{3}(1.5 + 4 \times 1.8 + 1.5) = 4.08 \text{ m}^2$$

Question 15 (a) (iii)

Criteria	Marks
• Correct explanation	1

Sample answer:

Simpson's rule uses a parabola through the 3 points. For the trapezoidal rule, the points are connected by straight lines. The roof hangs below the straight lines, and the parabola is above the straight lines. Hence, the area A is closer to the estimate given by the trapezoidal rule than that given by Simpson's rule.

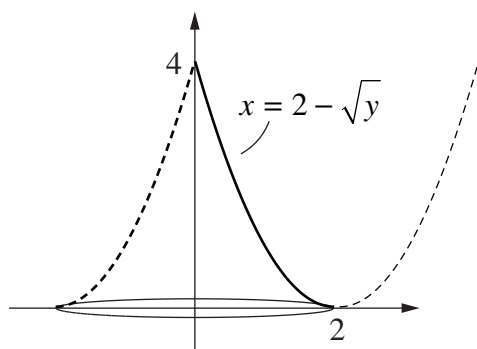


Alternative: For the trapezoidal rule, the top of the poles are connected by straight lines to form trapeziums. The exact area is less than the area of the trapeziums because the roof hangs between the poles. Since Simpson's rule gives a greater value than the trapezoidal rule, then the trapezoidal rule gives a better estimate.

Question 15 (b)

Criteria	Marks
• Correct solution	4
• Makes significant progress to get primitive of $(2 \pm \sqrt{y})^2$	3
• Obtains $V = \pi \int_0^a (2 \pm \sqrt{y})^2 dy$, or equivalent merit	2
• States $V = \pi \int_0^a x^2 dy$, for some value of a OR • Finds the y-intercept, or equivalent merit	1

Sample answer:



$$(x-2)^2 = y$$

$$x-2 = \pm\sqrt{y}$$

$$x = 2 \pm \sqrt{y}$$

But $x = 0$ when $y = 4$

$\therefore$ use the left branch

ie $x = 2 - \sqrt{y}$

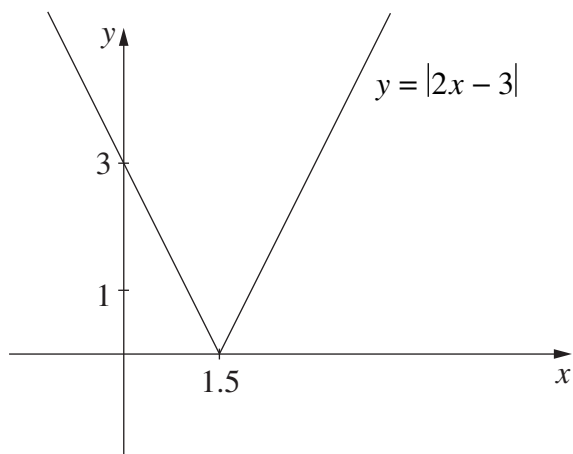
The volume of revolution is

$$\begin{aligned}
 V &= \pi \int_0^4 x^2 dy = \pi \int_0^4 (2 - \sqrt{y})^2 dy \\
 &= \pi \int_0^4 (4 - 4\sqrt{y} + y) dy \\
 &= \pi \left[4y - 4 \times \frac{2}{3} y^{\frac{3}{2}} + \frac{1}{2} y^2 \right]_0^4 \\
 &= \pi \left(16 - \frac{64}{3} + 8 - 0 \right) \\
 \therefore \text{volume} &= \frac{8\pi}{3} \text{ units}^3
 \end{aligned}$$

Question 15 (c) (i)

Criteria	Marks
Correct sketch	1

Sample answer:

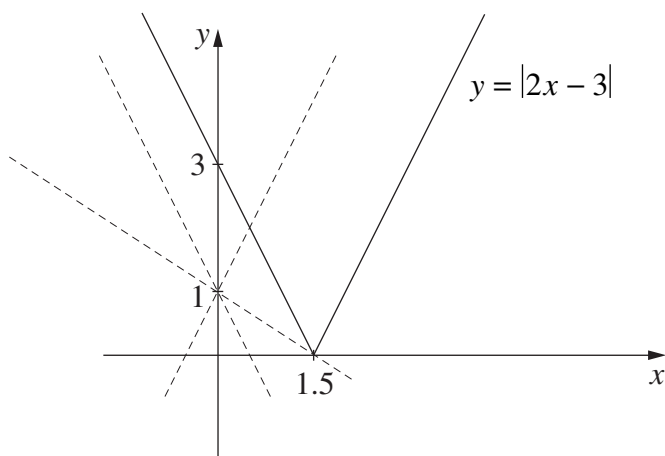

Question 15 (c) (ii)

Criteria	Marks
Correct solution	2
Finds at least one possible value for m, or equivalent merit	1

Sample answer:

From the graph:

$$m < -2 \quad \text{or} \quad m \geq 2 \quad \text{or} \quad m = -\frac{2}{3}$$



Question 15 (d) (i)

Criteria	Marks
• Correct answer	1

Sample answer:

As Pat starts, the probability for Pat to win the first throw is $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$

Question 15 (d) (ii)

Criteria	Marks
• Correct solution	2
• Finds probability that Pat wins in second game OR • Attempts to find probability Pat wins in the second game, and adds to answer from part (i)	1

Sample answer:

Pat winning on the second throw means both lose in the first throw and Pat wins on the next.

That probability is $\frac{35}{36} \times \frac{35}{36} \times \frac{1}{36}$.

The probability that Pat wins on the first or second throw:

$$\frac{1}{36} + \frac{35}{36} \times \frac{35}{36} \times \frac{1}{36} = \frac{2521}{46656} = 0.054$$

Question 15 (d) (iii)

Criteria	Marks
• Correct solution	2
• Finds probability that Pat wins in the third game, or equivalent merit	1

Sample answer:

For Pat to win on the third throw, probability is

$$\left(\frac{35}{36} \times \frac{35}{36} \times \frac{35}{36} \times \frac{35}{36} \times \frac{1}{36} \right)$$

∴ The probability that Pat eventually wins is

$$\begin{aligned} & \frac{1}{36} + \left(\frac{35}{36} \times \frac{35}{36} \times \frac{1}{36} \right) + \left(\frac{35}{36} \times \frac{35}{36} \times \frac{35}{36} \times \frac{35}{36} \times \frac{1}{36} \right) + \dots \\ &= \frac{1}{36} \left(1 + \left(\frac{35}{36} \right)^2 + \left(\frac{35}{36} \right)^{2 \times 2} \times \left(\frac{35}{36} \right)^{2 \times 3} + \dots \right) \end{aligned}$$

(This forms an infinite *GP* with $a = 1$, $r = \left(\frac{35}{36} \right)^2$ and $S_{\infty} = \frac{a}{1-r}$)

$$\begin{aligned} &= \frac{1}{36} \left(\frac{1}{1 - \left(\frac{35}{36} \right)^2} \right) \\ &= \frac{1}{36} \left(\frac{1}{1 - \frac{1225}{1296}} \right) \\ &= \frac{36}{71} \end{aligned}$$

Question 16 (a)

Criteria	Marks
• Correct solution	3
• Finds primitive of $f'(x)$ and solves $f'(x) = 5$, or equivalent merit	2
• Finds a primitive of $f'(x)$ OR • Solves $f'(x) = 5$, or equivalent merit	1

Sample answer:

Slope of tangent is given by $f'(x) = 4x - 3$.

Slope of line $y = 5x - 7$ is 5.

For the line to be tangent to $y = f(x)$

$$4x - 3 = 5$$

$$\therefore x = 2$$

When $x = 2$

$$y = 5x - 7$$

$= 3$ So the point $(2,3)$ lies on the graph $y = f(x)$ and on the tangent.

$$\text{Now } f(x) = \int (4x - 3) dx$$

$$= \frac{4x^2}{2} - 3x + C$$

Since $(2,3)$ lies on the graph

$$\begin{aligned} 3 &= f(2) \\ &= 2 \times (2)^2 - 3(2) + C \\ &= 2 + C \end{aligned}$$

so $C = 1$

$$\text{Hence } f(x) = 2x^2 - 3x + 1$$

Question 16 (b) (i)

Criteria	Marks
• Correct answer	1

Sample answer:

When $t = 0$, $N = 375 - e^0$

There were 374 trout.

Question 16 (b) (ii)

Criteria	Marks
• Correct solution	1

Sample answer:

Solve

$$N = 375 - e^{0.04t} = 0$$

$$e^{0.04t} = 375$$

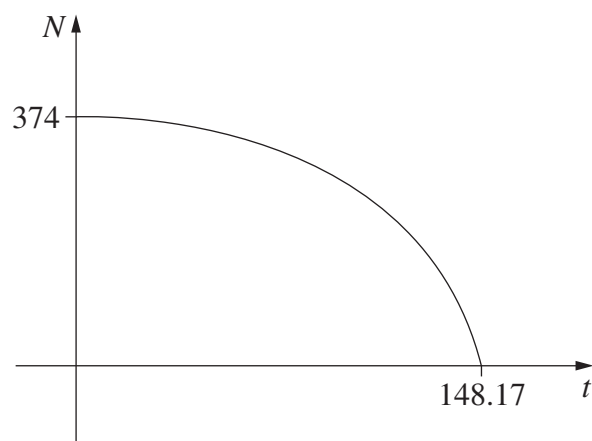
$$0.04t = \ln 375$$

$$t = \frac{\ln 375}{0.04} = 148.17$$

Question 16 (b) (iii)

Criteria	Marks
• Correct sketch	1

Sample answer:



Question 16 (b) (iv)

Criteria	Marks
• Correct solution	3
• Finds $0.2e^{0.02t} = 0.04e^{0.04t}$, or equivalent merit	2
• Finds $P = 10e^{0.02t}$ OR • States $\left \frac{dP}{dt} \right = \left \frac{dN}{dt} \right $, or equivalent merit	1

Sample answer:

P solves the equation $\frac{dP}{dt} = 0.02P$

So $P(t) = 10e^{0.02t}$, as $P(0) = 10$

For the rates to be equal,

$$\left| \frac{dN}{dt} \right| = \left| \frac{dP}{dt} \right|$$

ie

$$0.04e^{0.04t} = 0.02P(t)$$

$$= 0.2e^{0.02t}$$

$$e^{0.02t} = \frac{0.2}{0.04}$$

$$= 5$$

$$t = \frac{\ln 5}{0.02}$$

$$= 80.47$$

Question 16 (b) (v)

Criteria	Marks
• Correct solution	2
• Finds quadratic equation for $e^{0.02t}$, or equivalent merit	1

Sample answer:

$$P(t) = N(t)$$

$$10e^{0.02t} = 375 - e^{0.04t}$$

$$e^{0.04t} + 10e^{0.02t} - 375 = 0$$

$$\left(e^{0.02t}\right)^2 + 10e^{0.02t} - 375 = 0$$

$$\left(e^{0.02t} + 25\right)\left(e^{0.02t} - 15\right) = 0$$

$$\text{Hence } e^{0.02t} = -25 \text{ or } 15$$

Exponentials are positive, so

$$e^{0.02t} = 15$$

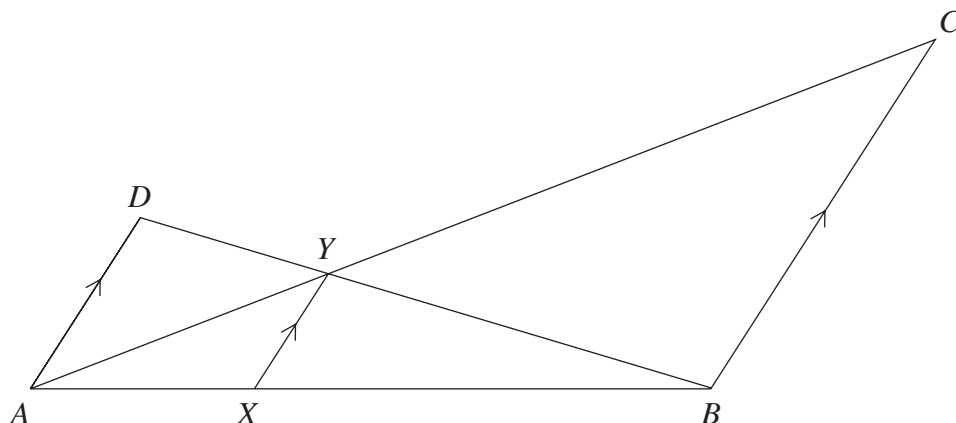
$$t = \frac{\ln 15}{0.02} = 135.4$$

The number of carp and trout are equal in the 136th month.

Question 16 (c) (i)

Criteria	Marks
• Correct proof	2
• Uses properties relating to parallel lines	1

Sample answer:



In $\triangle ABC$ and $\triangle AXY$

$\angle XAY$ is common

$\angle BCA = \angle XYA$ (corresponding angles $XY \parallel BC$)

Hence $\triangle ABC$ is similar to $\triangle AXY$ (equiangular)

Question 16 (c) (ii)

Criteria	Marks
• Correct proof	2
• Obtains a ratio involving XY and AX or BX , or equivalent merit	1

Sample answer:

By similarity of $\triangle ABC$ and $\triangle AXY$:

$$\frac{AX}{AB} = \frac{XY}{BC} \text{ (corresponding sides in similar triangles are in proportion).}$$

Now, $\triangle ABD$ is similar to $\triangle XBY$ (using similar argument used in part (i)). Hence

$$\frac{BX}{BA} = \frac{XY}{AD}$$

Add the two identities:

$$\frac{AX}{AB} + \frac{BX}{BA} = \frac{XY}{BC} + \frac{XY}{AD}$$

$$\frac{AB}{AB} = XY \left(\frac{1}{BC} + \frac{1}{AD} \right) \text{ (since } AX + BX = AB \text{)}$$

$$\frac{1}{XY} = \frac{1}{AD} + \frac{1}{BC}$$

Mathematics

2013 HSC Examination Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	1.4	P3, P4
2	1	5.3, 6.1	P5
3	1	1.4, 4.1	P5
4	1	8.9, 13.5	H5
5	1	3.3	H5
6	1	13.3	H5
7	1	9.5	P5
8	1	10.1, 10.4	H6
9	1	12.2, 12.3	H3
10	1	14.1	H5

Section II

Question	Marks	Content	Syllabus outcomes
11 (a)	1	1.1	P3
11 (b)	2	8.2	P3, P4
11 (c)	2	8.8, 13.5	P7, H5
11 (d)	2	8.8, 12.5	P7, H5
11 (e)	2	12.5	H5, H8
11 (f)	3	12.5	H5, H8
11 (g)	3	4.4	P4
12 (a)	2	10.4	H5
12 (b) (i)	2	6.2	P4
12 (b) (ii)	1	6.5	P4
12 (b) (iii)	1	6.5	P4
12 (b) (iv)	2	2.3, 6.7	P4
12 (c) (i)	2	7.1, 7.2	H5
12 (c) (ii)	2	7.2, 7.5	H5
12 (c) (iii)	3	7.1	H5
13 (a) (i)	2	5.2, 13.1	P4, H5
13 (a) (ii)	2	13.3	H5
13 (b) (i)	1	1.4, 9.2	P4

Question	Marks	Content	Syllabus outcomes
13 (b) (ii)	3	11.2, 11.4	H8
13 (c)	2	13.1	H5
13 (d) (i)	2	7.5	H4, H5
13 (d) (ii)	1	7.5	H4, H5
13 (d) (iii)	2	7.5	H4, H5
14 (a) (i)	1	14.3	H4, H5
14 (a) (ii)	1	14.3	H4, H5
14 (a) (iii)	2	14.3	H4, H5
14 (a) (iv)	2	14.3	H4, H5
14 (b) (i)	2	10.6	H4
14 (b) (ii)	3	10.6	H4, H5
14 (c)	3	5.3, 5.5, 13.2	P4, H5
14 (d)	1	11.4	H8
15 (a) (i)	1	11.3	H8
15 (a) (ii)	1	11.3	H8
15 (a) (iii)	1	11.3	H8
15 (b)	4	11.4	H8
15 (c) (i)	1	4.2	P5
15 (c) (ii)	2	1.2, 1.4, 4.2	P4, P5
15 (d) (i)	1	3.1	H5
15 (d) (ii)	2	3.2, 3.3	H5
15 (d) (iii)	2	3.2, 7.3, 7.5	H5
16 (a)	3	8.6, 10.8	P6, H8
16 (b) (i)	1	14.2	H3
16 (b) (ii)	1	14.2	H3
16 (b) (iii)	1	14.2	H3
16 (b) (iv)	3	14.2	H3
16 (b) (v)	2	14.2	H3
16 (c) (i)	2	2.5	H5
16 (c) (ii)	2	2.5	H5